

Classification or Regression?

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can identify a task from the meaning of its target.

I can distinguish category agreement from numerical error.

I can specify outputs and checks that match a user need.

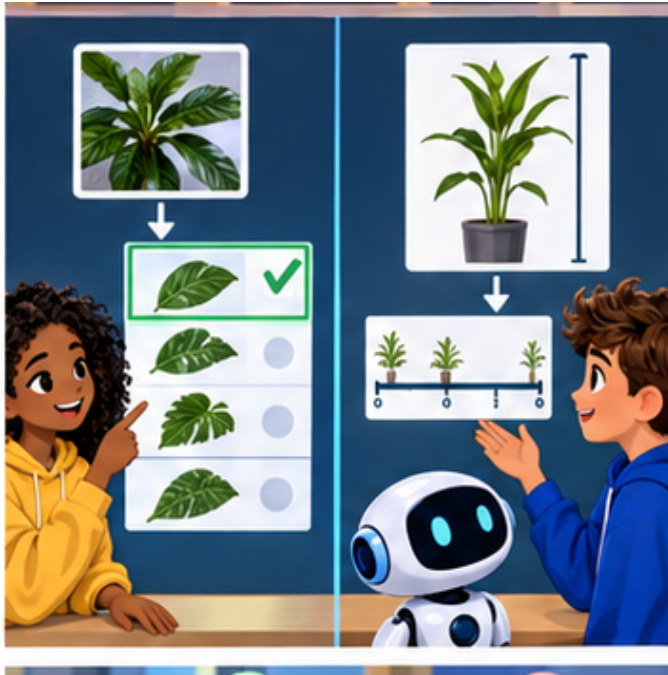
Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Classification predicts a category; regression predicts a numerical quantity. The target—not the input format—determines the task type.



LEARNING TARGET System A predicts class codes 0=oak,1=maple. System B predicts leaf area in cm^2 . Name the task types and explain why code 1 is not 'one more species' than code 0.

Builder name and experiment date:

Readiness for the connected explanation

At dispatch, may the observed arrival duration be used as an input to predict that same trip duration? Explain.

This week's end goal

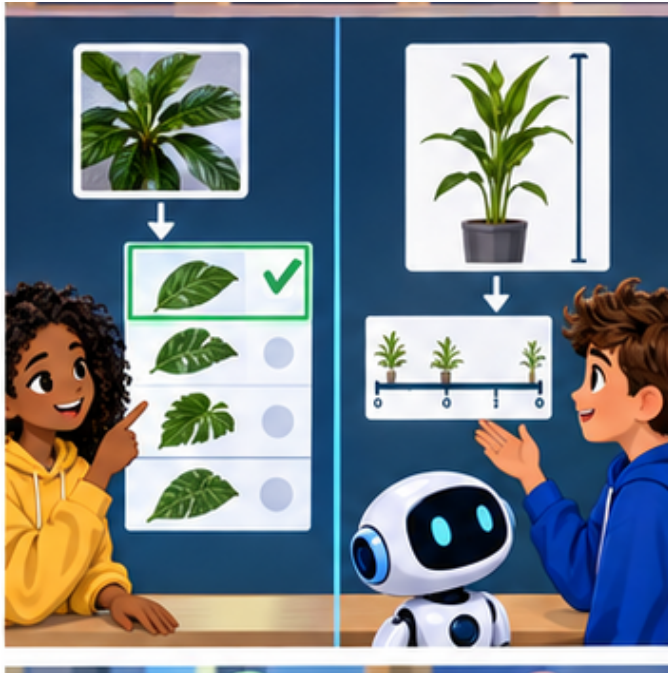
By the end, I can explain classification or regression? using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use classification or regression? to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

At robot dispatch, we can ask two questions about the same fictional trip. Will it take more than 10 minutes? Or how many minutes will it take? The first asks for a category. The second asks for an amount with units. The target question determines the task, even if both models would use the same dispatch-time inputs.

Question · Target meaning · Task

Question: Will the trip exceed 10 minutes?

Target meaning: 0 for no and 1 for yes

Task: Classification

Question: How long will the trip take?

Target meaning: Elapsed duration in minutes

Task: Regression

Question: Which route type will be used?

Target meaning: Indoor or outdoor

Task: Classification

Classification predicts a category from a defined set. Regression estimates a numerical quantity for which differences have meaning. A category may be coded with numbers: 0 and 1 here are names for no and yes, not travel times. Conversely, a duration recorded as a whole number of minutes can still be a regression target. Decimal points do not decide the task.

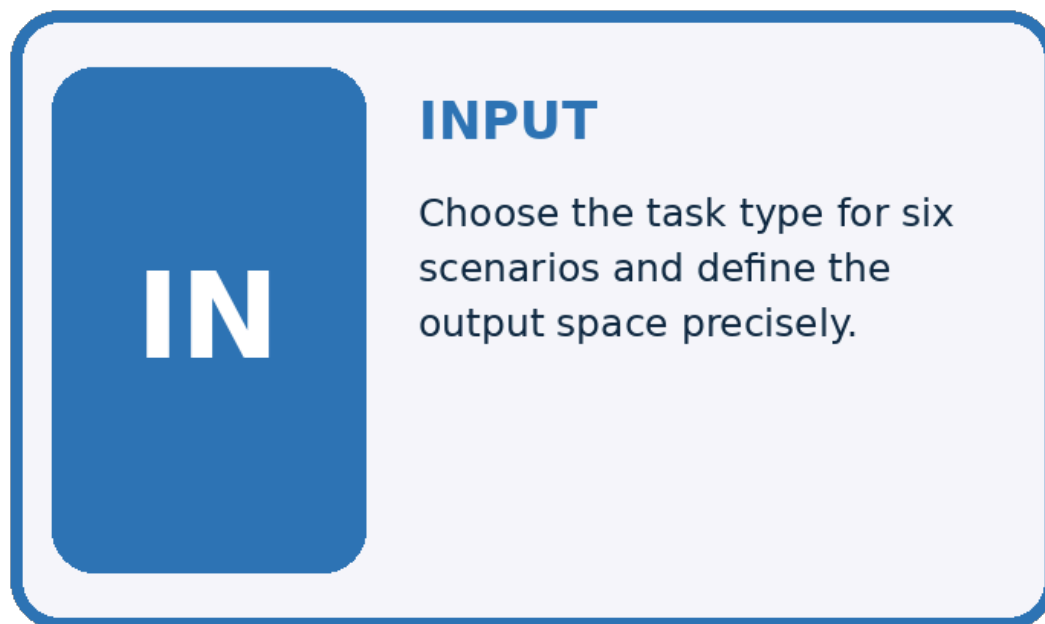
Keep Week 2's prediction time: route length and the saved queue may exist at dispatch, while the observed duration is available later for training or evaluation. Do not feed that observed duration into an earlier dispatch prediction.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.





ML

MODEL

Classification predicts a category; regression predicts a numerical quantity. The target—not the input format—determines the task type.



OUT

EVIDENCE

Choose the task type for six scenarios and define the output space precisely.

QUESTION Choose the task type for six scenarios and define the output space precisely.

MODEL IDEA Classification predicts a category; regression predicts a numerical quantity. The target—not the input format—determines the task type.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Predict plant species from a photo: target is a category, so classification. Predict plant height next week: target is centimeters, so regression.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

The following predictions are supplied for practice; no model is fitted here. Each estimate was written before revealing its matching observed duration. We compare duration error with a category decision made from that estimate. Predict slow when the estimated duration is greater than 10 minutes; exactly 10 is not slow. A classifier could also predict categories directly, but this example uses a threshold on estimates.

```
actual_minutes = [8, 10, 14]
estimated_minutes = [9, 11, 12]

correct = 0
errors = []

for actual, estimate in zip(actual_minutes, estimated_minutes):
    actual_slow = int(actual > 10)
    predicted_slow = int(estimate > 10)

    correct = correct + int(actual_slow == predicted_slow)

    errors.append(abs(estimate - actual))

print(correct, len(actual_minutes))

print(errors)
```

zip pairs the lists in order. They must have equal lengths and refer to the same trips; otherwise this loop is not a valid evaluation. abs gives the distance from zero, so both a one-minute overestimate and a one-minute underestimate have absolute error 1. The output is 2 3 followed by [1, 1, 2]. The mean absolute error is $(1 + 1 + 2) / 3 = 4/3$ minutes, about 1.33 minutes.

Read both kinds of evidence

Actual and estimate · Duration error · Category match

Actual and estimate: 8 and 9 minutes

Duration error: 1 minute

Category match: No and no match

Actual and estimate: 10 and 11 minutes

Duration error: 1 minute

Category match: No and yes differ

Actual and estimate: 14 and 12 minutes

Duration error: 2 minutes

Category match: Yes and yes match

All three rows use the same greater-than-10 boundary. A small duration error can cross that boundary; a larger error can stay on the same side.

Connect the picture and the explanation

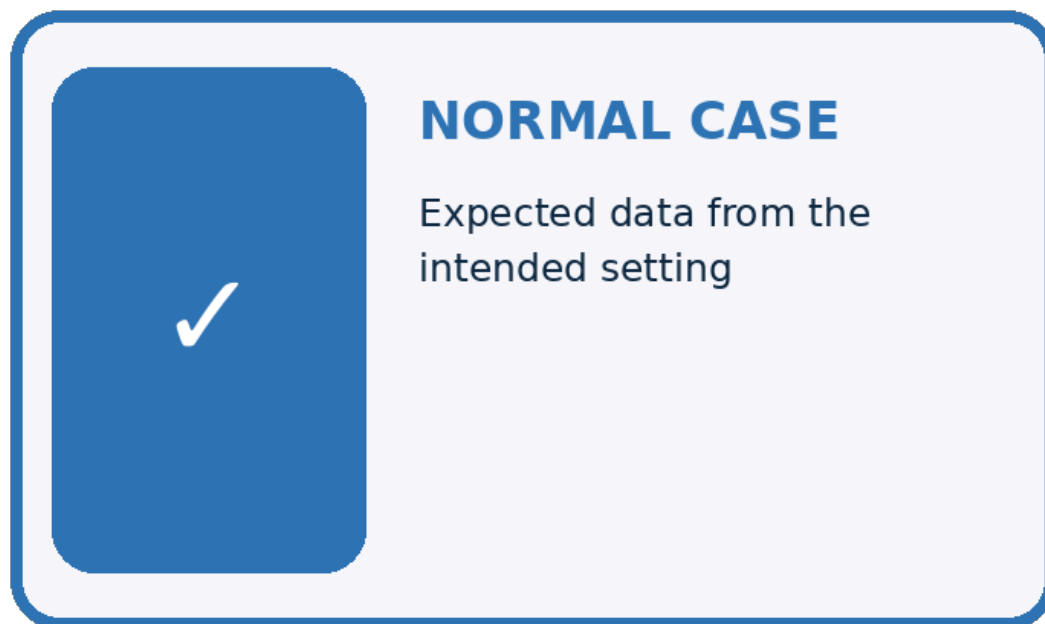
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





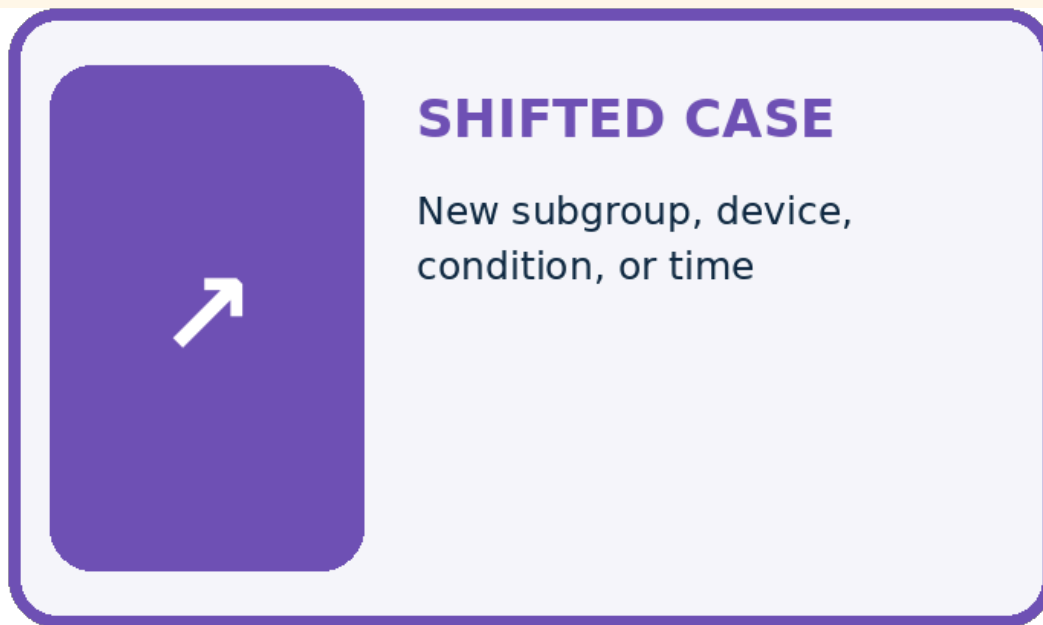
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Change the category boundary to strictly greater than 7 cm.

Prediction: Actual and estimated categories both become short, short, tall: 3/3 correct.

Reason: The numerical estimates did not improve; the decision definition changed.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

CLASSIFICATION

predict a category

2

REGRESSION

predict a number

3

CLASS

one allowed category

4

CONTINUOUS TARGET

number that can vary across a range

Term · Precise meaning · My ML example

Term: classification

Precise meaning: predict a category

My ML example: _____

Term: regression

Precise meaning: predict a number

My ML example: _____

Term: class

Precise meaning: one allowed category

My ML example: _____

Term: continuous target

Precise meaning: number that can vary across a range

My ML example: _____

MISCONCEPTION TO REPAIR A photograph always means classification and a table always means regression.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Classification

Meaning: Predicting a category from a defined set

Word: Regression

Meaning: Estimating a numerical quantity with meaningful differences

Word: Threshold

Meaning: A boundary used to turn a value into a decision

Word: Absolute error

Meaning: The nonnegative distance between an estimate and an observation

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 3 • WORKED EVIDENCE

- 1 Predict plant species from a photo: target is a category, so classification.
- 2 Predict plant height next week: target is centimeters, so regression.
- 3 Predict low/medium/high growth: categories, even though ordered.
- 4 State allowed classes or numerical unit before selecting a model and metric.

STEP 1 Predict plant species from a photo: target is a category, so classification.

STEP 2 Predict plant height next week: target is centimeters, so regression.

STEP 3 Predict low/medium/high growth: categories, even though ordered.

STEP 4 State allowed classes or numerical unit before selecting a model and metric.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

The 14-minute trip has the largest duration error in our example, yet its slow category is correct. The 10-minute trip has only a one-minute error but the category is wrong. These checks answer different questions. Report 2 correct out of 3 category decisions separately from a mean absolute error of about 1.33 minutes. Do not compare a fraction directly with minutes.

A trip actually takes 10 minutes and its estimate is 10.5 minutes. At the greater-than-10 boundary, give the actual category, predicted category and absolute duration error. Is the category decision correct?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Classification or Regression?

Distinguish predicting growth height from predicting a growth category.

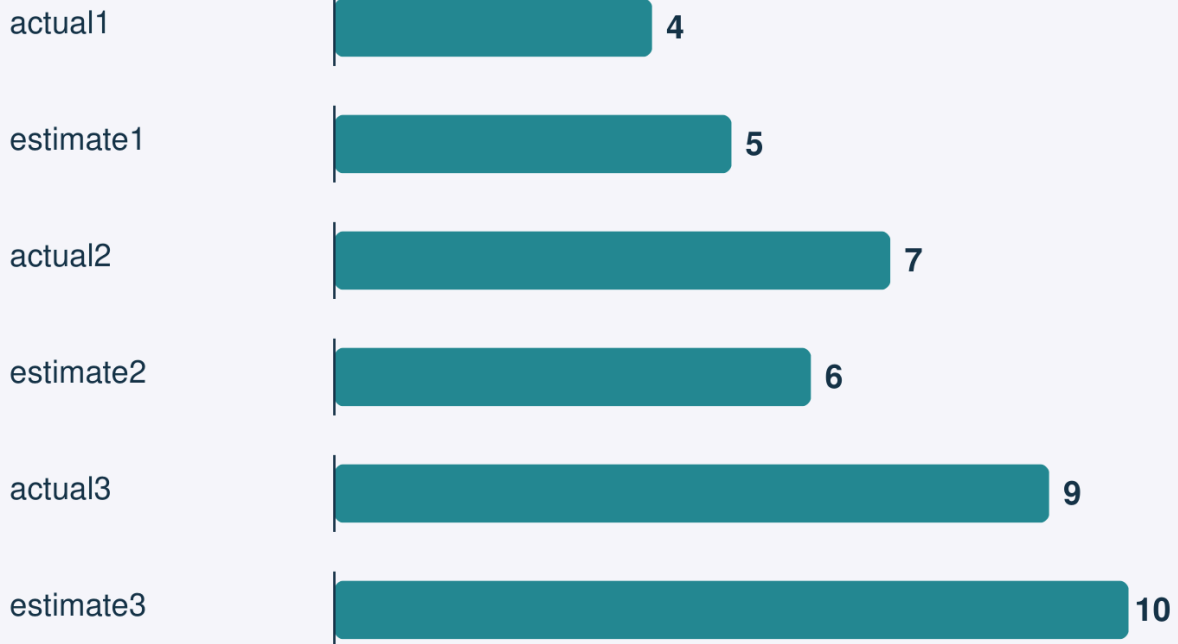
The facts we will use

Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Actual and estimated heights (paired order)

cm



Bar length shows the supplied value; the marked baseline is zero.

Classification or Regression?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regression errors

Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.



2. Actual categories

Actual classes are short, tall, tall.



3. Predicted categories

Estimated classes are short, short, tall: 2/3 correct.

CHANGE ONE THING

Change the category boundary to strictly greater than 7 cm.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Regression errors

What happens and why: Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.

Step: 2. Actual categories

What happens and why: Actual classes are short, tall, tall.

Step: 3. Predicted categories

What happens and why: Estimated classes are short, short, tall: 2/3 correct.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Estimated classes are short, short, tall: 2/3 correct. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Regression error and category accuracy measure different targets; neither alone covers every use consequence.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: A photograph always means classification and a table always means regression. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause / next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A photograph always means classification and a table always means regression.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A task specification should name the prediction moment, permitted inputs, target meaning and how predictions will be checked. Classification checks compare category decisions with observed categories. Regression checks measure numerical error with stated units. Later weeks examine additional metrics and why the consequences of different errors matter.

A correct category does not imply an exact quantity, and low numerical error does not guarantee every threshold decision is right. Three supplied examples establish only these calculations. They do not establish a trained model's reliability or show that a robot should act without review.

A team needs an estimated arrival duration in minutes. State the task type and a suitable error check. Explain why reporting only the fraction of correct slow or not-slow decisions leaves its main question unanswered.

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A learner says that predicting route type is regression because indoor is encoded as 1 and outdoor as 2. The numbers are category codes: their difference does not measure travel time or another meaningful amount. Treat this as classification and judge the predicted category. Changing the codes must not change

the meaning of the task.

Another learner puts durations into short and long bins, then claims to predict exact minutes. Binning creates a classification target and loses within-bin detail. If the user needs minutes, keep a quantity target and evaluate error in minutes. If the user needs a threshold decision, define the boundary and relevant consequences. Neither task is automatically better.

Practise with less help: Trace both checks

Use workbook W2 for the connected example “Classification and regression”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Classification and regression. Proposed explanation: “Numbers always mean regression”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Did regression MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Task-Type Clinic: classify six problem statements, rewrite vague targets, and match each to a useful metric.

1. Write the hypothesis and expected evidence before running.
2. Task-Type Clinic: classify six problem statements, rewrite vague targets, and match each to a useful metric.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: classification, regression, class, continuous target.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A photograph always means classification and a table always means regression.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER

Confirm whether Plant Classifier predicts a health category or a numerical severity score; document why.

THIS WEEK'S CHECKPOINT Confirm whether Plant Classifier predicts a health category or a numerical severity score; document why.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

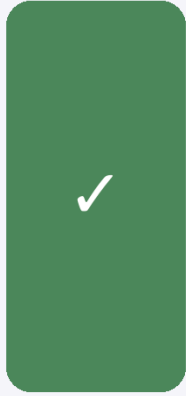
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Choose the task type for six scenarios and define the output space precisely.

REQUIRED PROOF Choose the task type for six scenarios and define the output space precisely.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Python and Table Readiness.

13. Study a complete case

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Task A predicts healthy/spotted; B predicts next week's height in cm.

The question

Choose task and output types.

Worked explanation

A is classification with categorical output; B is regression with numeric output and units. Numeric class codes do not make classification regression.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Choose the task by the meaning of the output

Evidence or stage · Value or result

Evidence or stage: Healthy or spotted

Value or result: Classification, category

Evidence or stage: Height next week

Value or result: Regression, numeric cm

Evidence or stage: Class code 0 or 1

Value or result: Still a category

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and output types.

A is classification with categorical output; B is regression with numeric output and units. Numeric class codes do not make classification regression.

14. Try a new case independently

Independent case: Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes. Write actual and predicted category lists, count correct categories, calculate each absolute error and the mean error in minutes. State what each result measures.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is classification and which is regression?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Confirm whether Plant Classifier predicts a health category or a numerical severity score; document why.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

System A predicts class codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.